LEARNER GUIDE

Can I write the line?

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

You have already met the two numbers that control a straight line on their own: the y-intercept c (where the line crosses the y-axis) and the gradient m (how steep it is, and which way it leans). This lesson brings them together. Once you know c and m, you can write the whole line as an equation: y = mx + c. You will play detective — reading c and m off a line — then build a line to match a target by setting the two dials yourself, spot the classic slips in other people's equations, and turn real situations into equations. The question you are exploring is: how do the gradient and the intercept become an equation?

What you are learning to notice

- That a straight line is pinned down by just two numbers — its intercept c and its gradient m.
- That the equation is y = mx + c: the gradient multiplies x, and the intercept is added on.
- That c is where the line crosses the y-axis (the value of y when x = 0), and m is the 1-across step.
- That both numbers keep their signs — a falling line has a negative m, and a line can have a negative c.

Moving through it, one step at a time

In Line Detective, read the two clues from the line — where it crosses the y-axis (c) and its gradient (m) — then write them into y = mx + c and press Reveal. In the Equation Builder, a faint target line is waiting: move the c slider to set where your line crosses, and the m slider to set its tilt, until your line lands on the target — and watch the equation build itself. In Right or slip?, decide whether someone's equation matches the line (watch for m and c swapped, or a lost minus sign). Then write the equation for a real situation with a starting value and a steady rate. There is no rush; use the coordinate grid on the Scratchpad to read c and m for yourself.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading the y-intercept from a graph, finding a gradient (the 1-across step), and knowing which number in y = mx + c is which), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.

- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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